

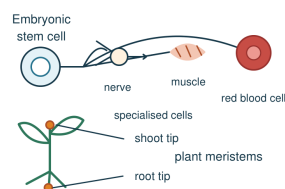


I am learning to explain how cells divide and then take on different jobs for growth and repair.

Words to use

Word	Meaning	Example	My language
differentiation	a cell becoming specialised for a particular job	A developing skin cell differentiates to do a job in skin.	
stem cell	a cell that can divide and form specialised cell types	Some adult stem cells make replacement blood cells.	
meristem	a plant region whose cells divide to make new cells	A root tip contains a meristem.	
growth	an increase in size, including by making more cells	A shoot grows as its cells divide and elongate.	

Model



An embryonic stem cell can give rise to different specialised cells; meristems at growing plant tips produce new cells. Simplified; not to scale.

In skin, mitosis adds cells; some stem-cell descendants differentiate for a tissue job. At a shoot tip, meristem cells divide, then elongate and differentiate.

How I know

I can distinguish making more cells from cells taking on a specialised job.

I can compare animal growth with plant growth.

I can identify a role for an animal stem cell or a plant meristem.



You can plan in any language. Speak before you write: Speaking: “A plant also ..., whereas an animal ...”